CS & IT ENGINEERING



Discrete Mathematics

Graph Theory

DPP-03 Discussion Notes



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[MCQ]

#Q. Which of the following options is/are correct ?



A The chromatic number of a graph with at least 1 edge is at least 2.

B A graph is null graph if and only if its chromatic number is 1.

C For any graph G, chromatic number of G $\leq 1+\Delta(G)$.

D For any graph G, chromatic number of G $\leq |V(G)|$

[MCQ]

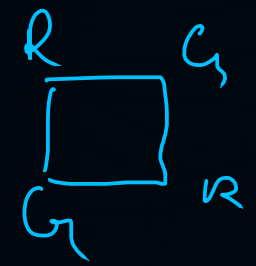
#Q. Which of the following options is/are correct ?

A. If chromatic number of graph G is 'm', then graph G is a m-partite graph, but it can never be (less than m)-partite.

B. If graph G is not a null graph and all the cycles of graph G are of even length, then chromatic number of G is 2.

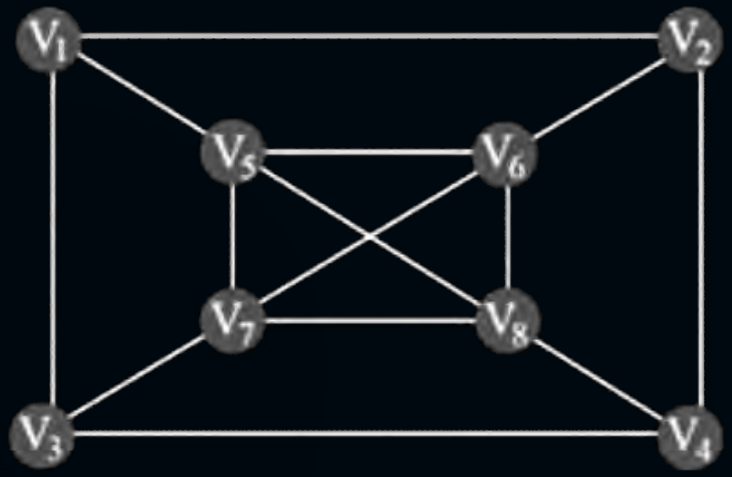
C. If graph G is not a null graph and all the cycles of graph G are of even length, then G is a bipartite graph.

D. A tree with at least one edge is always a bipartite graph.



[NAT]

#Q. Consider the following graph,



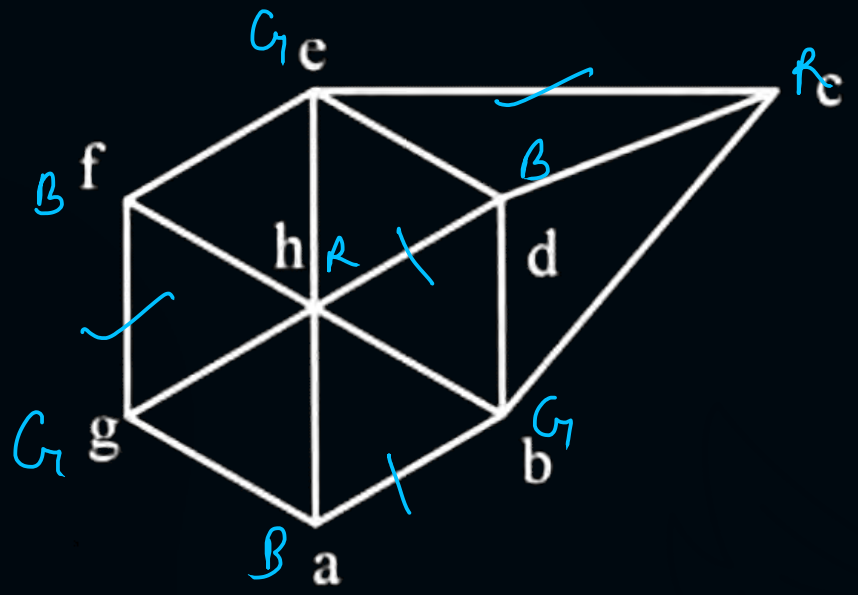
→ odd length cycle

(2)

Minimum number of edges that must be deleted from the graph such that the graph becomes a bipartite graph is_____

[MCQ]

#Q. For the graph G shown below



Chromatic number of G + matching number of G = _____.

3 + 4

A 6

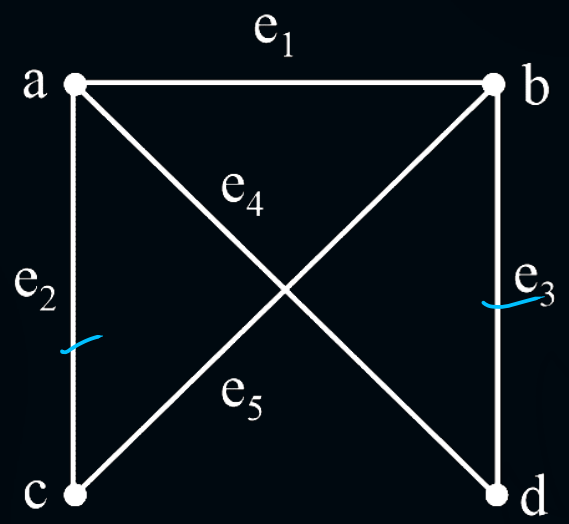
B 7

C 5

D 8

[NAT]

#Q. Number of maximal matching in the graph shown below is 3 ?



{e1}

{e2, e3}

{e4, e5}

[MCQ]

#Q. Which of the following is not True ?

A. Number of perfect matching in $k_{2n} = \frac{(2n)!}{2^n n!}$

B. Number of perfect matching in $K_{n,n} = n!$

C. Number of perfect matching in C_n (n is even)$= 2$

D. Number of perfect matching in $w_{2n} = 2n$

[MCQ]

#Q. Number of perfect matching in a tree with n – vertices (n ≥ 2)

A >1

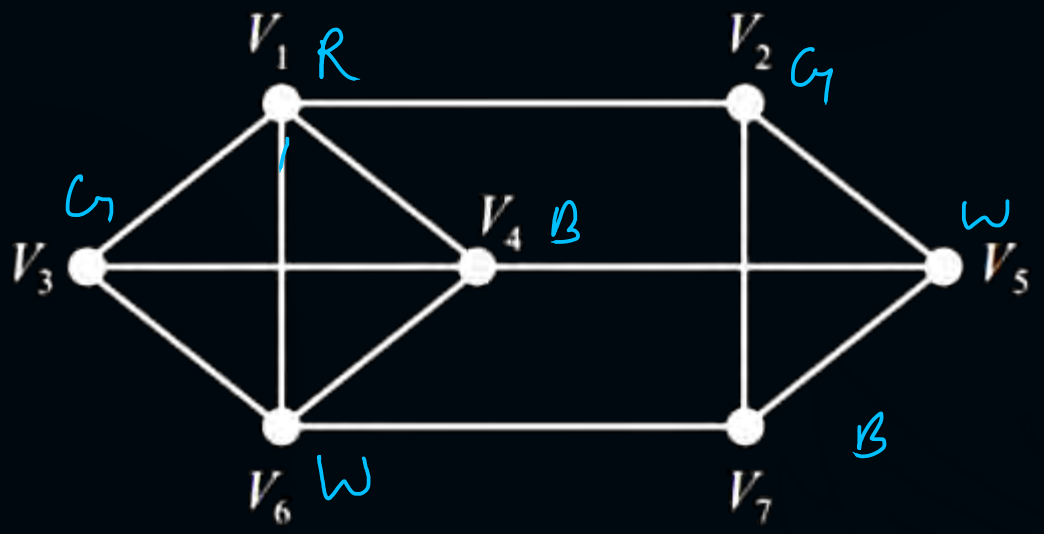
B ≥ 1

C < 1

D = 1

[NAT]

#Q. For the graph shown below, the chromatic number is 4.



[NAT]

#Q. Find the matching number of the following graph.



a b

c d

g h

3

[NAT]

#Q. Let G be a simple graph with 20 Vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is_____

$$VC + IS = n$$

$$\Rightarrow 8 + IS = 20$$

$$\Rightarrow IS = 12$$

[MCQ]

#Q. What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

A 5

B 4

C 3

D 2

THANK - YOU